

TRẦN HUỲNH ĐỨC TÀI

AI/MACHINE LEARNING ENGINEER

Ho Chi Minh City, Vietnam | +84338241002 | hdt.tran24100@gmail.com | [github](#) | [linkedin](#) | [portfolio](#)

Personal Statement

AI Engineer and Data Science graduate with research experience in Healthcare AI, Computer Vision, Edge AI, and Generative AI. Skilled in developing deep learning systems for physiological signal analysis, medical image segmentation, 3D reconstruction, and retrieval-augmented generation (RAG). Published author in healthcare AI research with experience translating research ideas into practical and deployable AI solutions. Seeking opportunities as an AI Research Engineer, Machine Learning Engineer, or Healthcare AI Engineer.

Education

Degree: Bachelor of Science in Data Science 2020 – 2025

International University - Vietnam National University, HCMC

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence, Data Mining, Statistical Learning

Work Experience

AI Research Assistant 01/2025 – Present

AI Lab – CSE, International University - HCMC, Vietnam

- Developed deep learning pipelines for stress recognition using wearable physiological signals.
- Designed CNN-LSTM architectures for temporal modeling and subject-independent classification.
- Investigated DDQN-based Active Learning strategies to improve data efficiency and model generalization.
- Built end-to-end preprocessing, training, and LOSO evaluation pipelines.
- Contributed to medical computer vision research involving wound segmentation, 3D reconstruction, and bio-printable patch generation.
- Participated in scientific writing, experiment design, and publication preparation.

Research Outcomes:

- Published a Springer LNNS paper on stress recognition achieving 95.1% LOSO accuracy.
- Co-authored a biomedical engineering paper accepted at BME11.
- Currently developing Edge AI solutions for healthcare monitoring systems.

Backend Developer Intern 10/2024 – 01/2025

Vitalify Asia Co., Ltd. - HCMC, Vietnam

- Developed backend services using NestJS and TypeORM.
- Implemented JWT-based authentication and authorization systems.
- Integrated third-party APIs for media processing workflows.
- Containerized applications using Docker and participated in AWS deployment pipelines.
- Contributed to CI/CD workflows and production deployment.

Publications

- Tran, H. D. T., et al. (2026), “A Hybrid Deep Learning Approach to Stress Detection: Integrating CNN-LSTM with Reinforcement Learning and Active Learning,” *Lecture Notes in Networks and Systems*, 1581, pp. 317–330: 2026.
- Nguyen, P.T.A., Tran, C., Tran, H.D.T., et al. (2026), “Automated Surface-to-Solid Conversion for Bio-Printable Patch Fabrication from Three-Dimensional Wound Surface Models”. Accepted at the 11th International Conference on Biomedical Engineering in Vietnam (BME11), 2026.

Projects

Automated Bio-Printable Wound Patch Generation 05/2026

- Developed an end-to-end medical imaging pipeline from segmentation to 3D reconstruction.
- Implemented EfficientNet-B4 U-Net segmentation and COLMAP reconstruction.
- Generated fabrication-ready STL patches with 23.3% bio-ink reduction.

- Accepted for publication at BME11.

Tools: Python, PyTorch, OpenCV, COLMAP, Open3D, Trimesh

Medical Research Assistant (RAG)

11/2025

- Built a Retrieval-Augmented Generation system for biomedical literature.
- Implemented FAISS retrieval and Cross-Encoder reranking.
- Developed a Streamlit-based interface supporting citation-aware answers.

Tools: Python, LangChain, FAISS, Gemini API, Streamlit

Human Stress Recognition from Wearable Device using Deep Learning

01/2025

- Developed a CNN-LSTM framework for stress classification using physiological signals.
- Proposed a DDQN + Active Learning framework to improve sample efficiency.
- Evaluated using LOSO validation on the WESAD dataset.
- Achieved 95.1% accuracy and published the work in Springer LNNS.

Tools: Python, TensorFlow, Keras, DDQN, NumPy, SciPy

Human Activity Recognition using Machine Learning

06/2024

- Designed a CNN-LSTM architecture to model spatial-temporal patterns in wearable sensor data.
- Performed data preprocessing and feature extraction for time-series signals.
- Evaluated model performance and optimised training strategies on the PPG-DaLiA dataset, improved classification accuracy from 48.3% to 64.3%, representing a 33.1% relative improvement over the baseline.

Tools: Python, TensorFlow, Pandas, NumPy, Scikit-learn

Web-based Interactive Visualization

06/2024

- Developed an interactive D3.js visualization platform for socio-economic datasets.
- Implemented ETL pipelines and dynamic data loading from GitHub.
- Built responsive dashboards featuring scatter plots and choropleth maps.

Tools: JavaScript, D3.js, HTML, CSS

Certifications and Licences

- **Generative AI Applications with RAG and LangChain** - IBM, Sep 2025
- **Artificial Intelligence Fundamentals** - IBM-SkillsBuild, Sep 2025
- **Generative AI - A Way of Life** - Analytics Vidhya, June 2025
- **Data Science Orientation** - Coursera, Sep 2023

Professional Relevant Skills

- **Programming:** Python, SQL, C++, Java
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, CNN, LSTM, CNN-LSTM, Transfer Learning, Reinforcement Learning (DDQN), Active Learning
- **Computer Vision:** OpenCV, Image Segmentation, EfficientNet, U-Net, COLMAP, Open3D, 3D Reconstruction
- **GenAI:** LangChain, RAG, FAISS, Cross-Encoder Reranking, Prompt Engineering
- **Deployment & MLOps:** Docker, AWS, REST APIs, Linux, Git, Streamlit
- **Data:** NumPy, Pandas, SciPy, ETL Pipelines
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Languages:** English (professional working proficiency)